PROBLEM SET 6, PART 1: TENSORS AND DIFFERENTIAL FORMS
DUE: DEC. 14

(1) [**Tedious multi-linear algebra**]
Prove Lemma 5.1.19 and Proposition 5.1.21.

(2) [**Decomposable tensors**]
A tensor $T \in V_1 \otimes V_2 \otimes \cdots \otimes V_k$ is called *decomposable* if there exists $v_1 \in V_1, \cdots, v_k \in V_k$ so that $T = v_1 \otimes \cdots \otimes v_k$. [Indecomposable tensors are known as *entangled states* in quantum information theory.]
   (a) Let v_1, v_2 be linearly independent vectors in V. Show that the 2-tensor $v_1 \otimes v_1 + v_2 \otimes v_2 \in V \otimes V$ is not decomposable.
   (b) Let V, W be vector spaces, with basis $\{v_i\}$ and $\{w_j\}$ respectively. Prove: A nonzero $T = \sum_{ij} a_{ij} v_i \otimes w_j \in V \otimes W$ is decomposable if and only if the coefficient matrix (a_{ij}) has rank 1.
   (c) [not required] For any vector space V, we let $\mathbb{P}(V) = V - \{0\}/\{v \sim \lambda v\}$ be the projective space of V. Prove: The map

$$\mathbb{P}(V_1) \times \cdots \times \mathbb{P}(V_k) \to \mathbb{P}(V_1 \otimes \cdots \otimes V_k), \quad ([v_1], \cdots, [v_k]) \mapsto [v_1 \otimes \cdots \otimes v_k]$$

   is an embedding. This is known as the (generalized) *Segre embedding*.
   (d) Suppose $k \geq 2$, and $\dim V_i \geq 2$ for each $1 \leq i \leq k$. Prove: Any decomposable k-tensor $T \in V_1 \otimes \cdots \otimes V_k$ lines in the image of the generalized Segre embedding above. Conclude that most k-tensors are indecomposable.

(3) [**Tensor product of vector spaces**]
Let V, W be finite dimensional vector spaces, with basis $\{v_i\}$ and $\{w_j\}$ respectively.
   (a) Prove: $V \otimes W^* \simeq L(W, V)$ as vector spaces. Then answer the following problems:
      (i) What is the trace of $T \in V \otimes V^* \simeq L(V, V)$?
      (ii) Give a conceptional proof of the fact that the contraction defined in Definition 5.1.5 is well-defined.
   (b) [not required] **[Universal Property]** Prove: There exists a blinear map $\varphi : V \times W \to V \otimes W$, so that the following *universal property* holds: for any vector space Z and any bilinear map $h : V \times W \to Z$, there exists a unique linear map $\tilde{h} : V \otimes W \to Z$ so that $h = \tilde{h} \circ \varphi$. Moreover, such a pair $(V \otimes W, \varphi)$ is unique up to isomorphism.

(4) [Lie derivative on differential forms] [Not required]
   (a) Read section 5.2.3.
   (b) Prove Proposition 5.2.11.

(5) [**Maxwell's equations**]

The famous *Maxwell's equations* in the theory of electromagnetism are of the form

$$\nabla \cdot \vec{B} = 0, \qquad \frac{\partial \vec{B}}{\partial t} + \nabla \times \vec{E} = 0,$$
$$\nabla \cdot \vec{E} = 4\pi\rho, \qquad \frac{\partial \vec{E}}{\partial t} - \nabla \times \vec{B} = -4\pi\vec{J},$$

where $\vec{E} = \langle E_x, E_y, X_z\rangle$ is the electricity field, $\vec{B} = \langle B_x, B_y, B_z\rangle$ is the magnetism field, ρ is the charge density, and $\vec{J} = \langle J_x, J_y, J_z\rangle$ is the current density. All functions, e.g. E_x, E_y, B_x, ρ, J_x, are functions of x, y, z and t.

Define the *Faraday 2-form* F on $\mathbb{R}^4$ (with coordinates t, x, y, z) as

$$\mathbf{F} = E_x dx\wedge dt + E_y dy\wedge dt + E_z dz\wedge dt + B_x dy\wedge dz + B_y dz\wedge dx + B_z dx\wedge dy,$$

and define the current 1-form to be

$$\mathbf{J} = \rho dt + J_x dx + J_y dy + J_z dz.$$

We also define the *Hodge star operators* on $\mathbb{R}^4$ (with respect to the *Minkowski metric*) to be the linear operators

$$* : \Omega^k(\mathbb{R}^4) \to \Omega^{4-k}(\mathbb{R}^4), \qquad 0 \le k \le 4,$$

so that on 2-forms one has $** = -1$ and

$$*(dx\wedge dt) = dy\wedge dz, \qquad *(dy\wedge dt) = -dx\wedge dz, \qquad *(dz\wedge dt) = dx\wedge dy,$$

while on 1-forms one has $dt\wedge(*dt) = dx\wedge(*dx) = \cdots = dt\wedge dx\wedge dy\wedge dz$, e.g.

$$*dt = dx\wedge dy\wedge dz, \quad *dx = -dt\wedge dy\wedge dz, \quad *dy = dt\wedge dx\wedge dz \quad \textit{etc.}$$

Prove: The system of Maxwell equations is equivalent to

$$d\mathbf{F} = 0, \qquad d*\mathbf{F} = 4\pi * \mathbf{J}.$$

(6) [**The canonical symplectic form on** $\mathbb{R}^{2n}$]

Denote by $(x^1,\cdots,x^n,y^1,\cdots,y^n)$ the coordinate functions on $\mathbb{R}^{2n}$. Let $\omega = \sum_{i=1}^{n} dx^i\wedge dy^i$.

(a) What is $d\omega$?
(b) What is $\omega^n = \omega\wedge\cdots\wedge\omega$ (wedge n times)?
(c) Find $\iota^*\omega$ for the following three cases:
  (i) $\iota : \mathbb{R}^{2n-2} \hookrightarrow \mathbb{R}^{2n}$ be the embedded submanifold of $\mathbb{R}^{2n}$ defined by $x^n = y^n = 0$.
  (ii) $\iota : \mathbb{R}^n \hookrightarrow \mathbb{R}^{2n}$ be the submanifold of $\mathbb{R}^{2n}$ defined by $y^1 = \cdots = y^n = 0$.
  (iii) $\iota : \mathbb{T}^n \hookrightarrow \mathbb{R}^{2n}$ be the submanifold of $\mathbb{R}^{2n}$ defined by $(x^i)^2 + (y^i)^2 = 1, 1 \le i \le n$.
(d) For any $f \in C^\infty(\mathbb{R}^{2n})$, find a vector field X_f so that $\iota_{X_f}\omega = df$.
(e) Let X_f be as above. Prove: $\mathcal{L}_{X_f} f = 0$ and $\mathcal{L}_{X_f}\omega = 0$.

(7) [**Involutive distributions revisited**]

(a) Let $\omega = ydx - dz$ be a smooth 1-form on $M = \mathbb{R}^3$. For each $p \in M$, let $\mathcal{V}_p := \ker(\omega|_{T_pM})$. Prove: $\mathcal{V}$ is a distribution of rank 2 on M. Is it integrable?

(b) Let $\mathcal{V}$ be a smooth distribution of rank k on M. Prove: For each $p \in M$, there exists a neighborhood U of p and smooth 1-forms $\omega_1, \cdots, \omega_{n-k}$ on U such that for each $q \in U$, $\mathcal{V}_q = \ker(\omega_1|_{T_qM}) \cap \cdots \cap \ker(\omega_{n-k}|_{T_qM})$.

Definition 0.1. We say a differential form $\eta \in \Omega^j(M)$ annihilates a smooth distribution $\mathcal{V}$ if $\eta(X_1, \cdots, X_j) = 0$ for all $X_i \in \mathcal{V}$.

(c) [Not required] Let $\mathcal{V}$ be a smooth distribution of rank k on M, and $\omega_1, \cdots, \omega_{n-k}$ are as described above. Prove: $\eta \in \Omega^j(M)$ annihilates $\mathcal{V}$ if and only if there exists $(j-1)$-forms $\alpha_1, \cdots, \alpha_{n-k}$ so that $\eta = \sum_{i=1}^{n-k} \omega_i \wedge \alpha_i$.

(d) [Not required] Prove: a smooth distribution $\mathcal{V}$ is involutive if and only if the following criteria is satisfied:

> If η is a 1-form that annihilates $\mathcal{V}$ on an open set U, then $d\eta$ also annihilates $\mathcal{V}$ on U.

(e) [Not required] Let $\omega_1, \cdots, \omega_{n-k}$ be as in (b). Prove: $\mathcal{V}$ is involutive if and only if there exists 1-forms α^i_j $(1 \le i, j \le n-k)$ so that

$$d\omega^i = \sum_{j=1}^{n-k} \omega^j \wedge \alpha^i_j, \qquad 1 \le i \le n-k.$$

(Ref: John Lee, *Introduction to Smooth Manifolds* page 493-495.)